

Data Governance Models for Higher Education Institutions

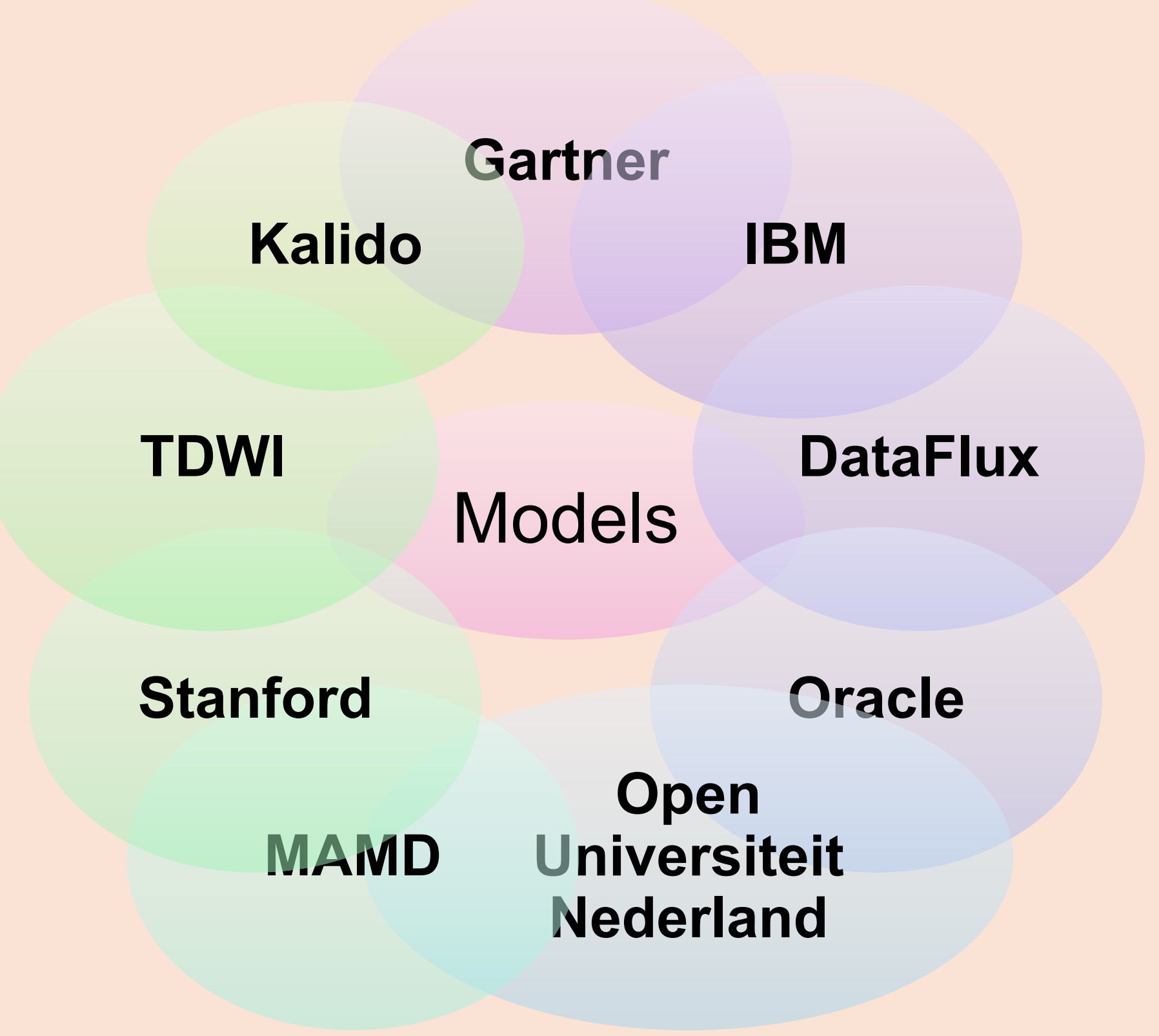
Nohema Pérez-España¹, Dra. Saiph Savage²,
Dr. Javier Ortiz-Hernández, ¹Dra. Yasmín Hernández¹
¹Tecnológico Nacional de México campus CENIDET, ²Northeastern University USA
d23ce174@cenidet.tecnm.mx, {javier.oh, yasmin.hp}@cenidet.tecnm.mx

Abstract

Higher Education Institutions (HEIs) manage vast amounts of data, yet most existing data governance models are not designed for their unique needs. This research identifies the strengths and weaknesses of current models and develops a new governance framework tailored specifically for Higher Education Institutions.

Relevant Literature

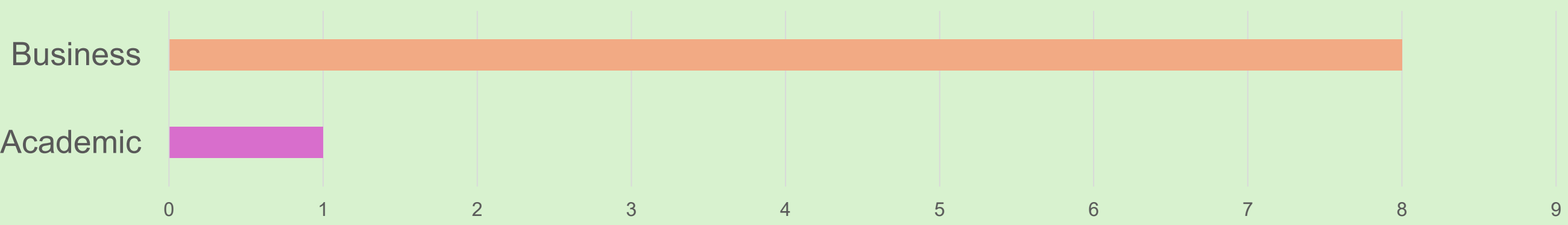
We connect with prior research that has developed metrics for evaluating and studying different data governance models (they evaluate the maturity of a given data governance model).



Problem Context

- Most models for assessing data governance maturity do not consider the academic context.

Context of models for assessing maturity



- The characteristics of these data governance models often fail to align with the needs of Higher Education Institutions (HEIs).

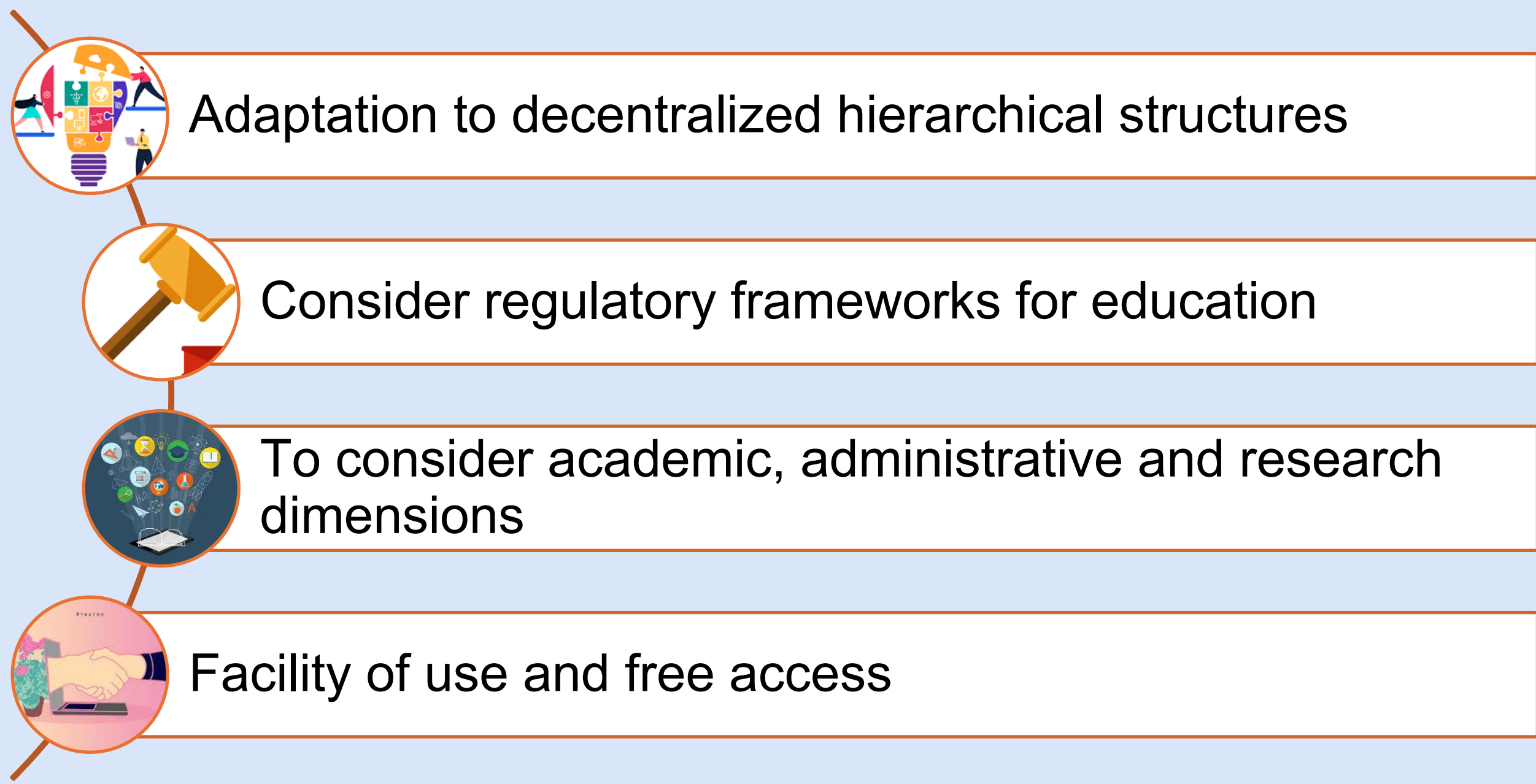


- This creates the need to create a new data governance model that more accurately reflects the specific context of Higher Education Institutions.

Methodology



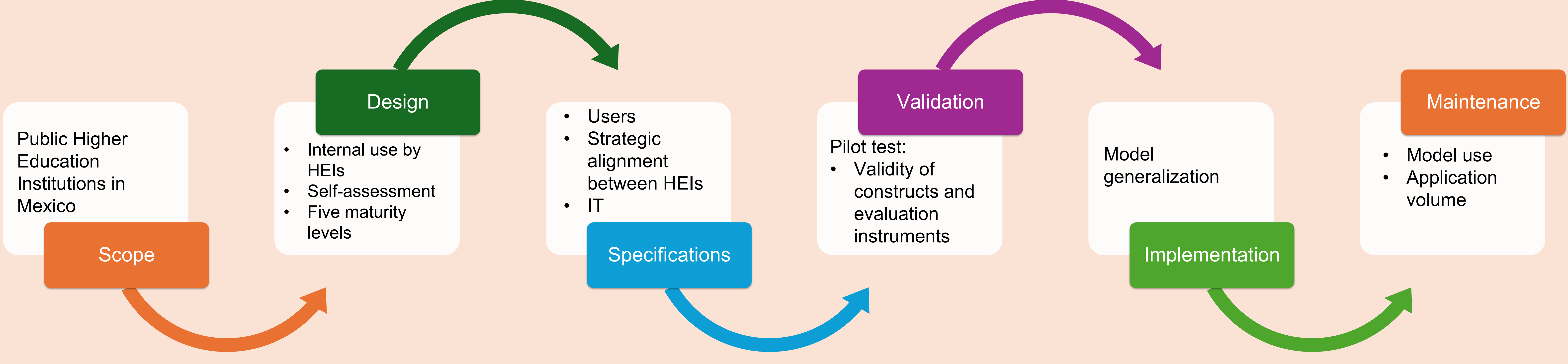
Data governance requirements in Higher Education



Identification of gaps in traditional governance models

Models	Gaps
Gartner	Consider regulatory and normative frameworks in the academic field.
IBM	Incorporate educational, ethical, and social dimensions.
Oracle	Adapt it to decentralized hierarchical structures.
Stanford	Consider ethical and regulatory aspects specific to the education sector.
DataFlux	Include academic indicators, not just technical ones
MAMD	Complement with regulatory frameworks specific to the academic sector.
TDWI	Incorporate cultural maturity metrics into the use of information.
Open Universiteit Nederland	Expand its institutional scope, including administrative and research functions.
Kalido	Adapt it to the academic context, considering institutional capabilities and available resources.

Design of a Maturity Model to assess data governance in Higher Education Institutions.



Conclusions.

Rigidity, generic focus, and limited adaptation to the academic context are some of the limitations of existing models for evaluating data governance maturity. Developing a model focused on the context of Higher Education Institutions in México will be enable more effective improvement implementations.

Technological Reflections

The instruments used to assess data governance maturity are mostly subjective and qualitative. One of the main technological challenges is to systematize the diagnostic process, reducing subjectivity by transforming qualitative data into quantifiable metrics.